🛡️ Task 1: Foundation & Environment Setup Report

Prepared by: Fatimah Adnan

Date: April 2026

Internship ID: APSPL2631283

---

1. Executive Summary

The objective of this task was to establish a secure, virtualized environment for penetration testing. This report documents the successful installation of an attacker machine (Kali Linux) and a vulnerable target (Metasploitable2), verified through network diagnostics and cryptographic integrity checks.

2. Lab Environment Configuration

The lab was built using a **Host-Only Networking** model to ensure that any security testing remains isolated from the physical host and the public internet.

- **Hypervisor:** VirtualBox 7.x
- **Attacker Node:** Kali Linux (IP: 192.168.56.101)
- **Target Node:** Metasploitable2 (IP: 192.168.56.102)
- **Network Isolation:** Host-Only Adapter (Subnet 192.168.56.0/24)

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:0c:74:a3
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe0c:74a3/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:9 errors:0 dropped:0 overruns:0 frame:0
          TX packets:30 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1818 (1.7 KB)  TX bytes:3924 (3.8 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:110 errors:0 dropped:0 overruns:0 frame:0
          TX packets:110 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:27753 (27.1 KB)  TX bytes:27753 (27.1 KB)

msfadmin@metasploitable:~$ _
```

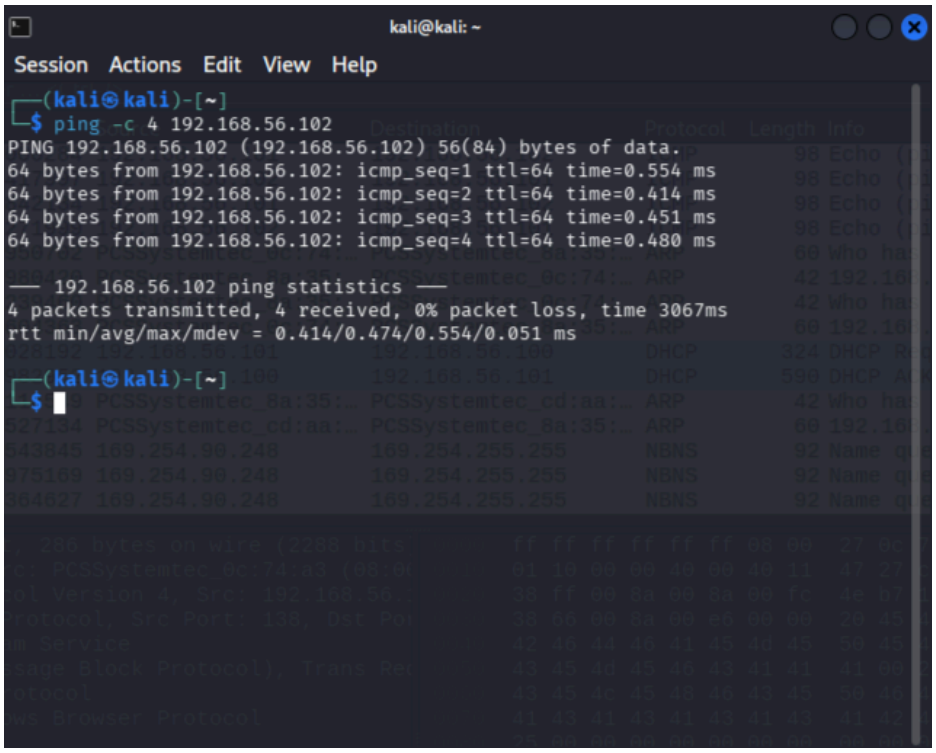
3. Network Verification (The "Availability" Pillar)

To confirm the target was reachable, a series of Layer 3 and Layer 4 tests were conducted.

Step 1: ICMP Connectivity (Ping)

I executed a `ping` command from Kali to the Target to verify basic connectivity.

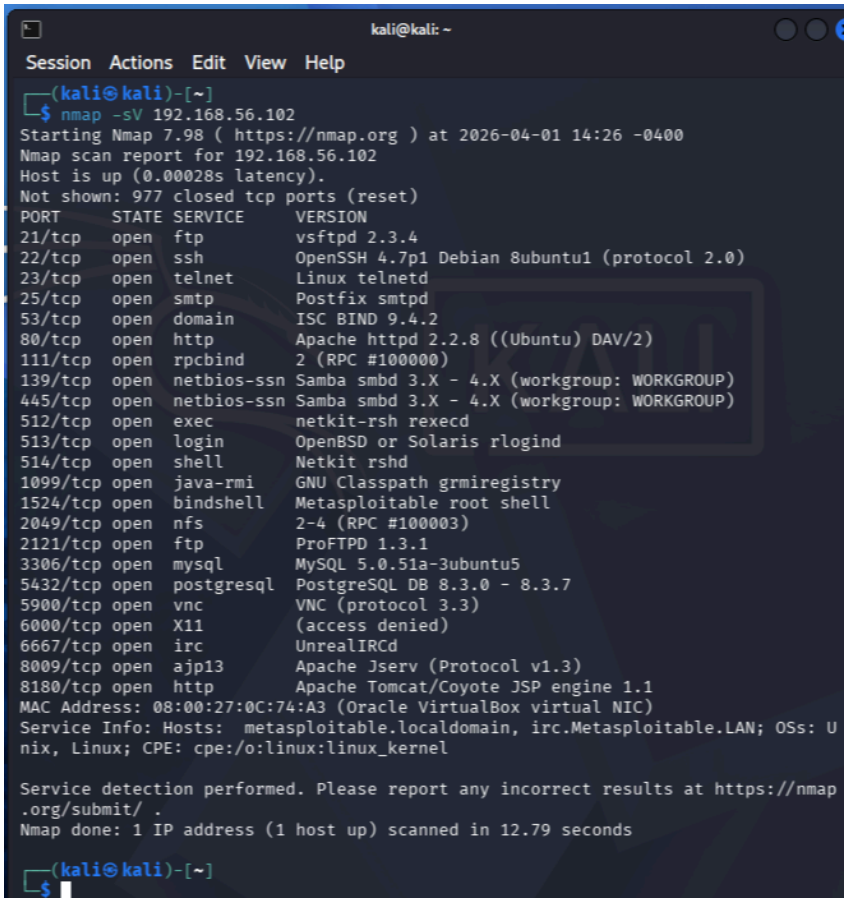
- **Command:** `ping -c 4 192.168.56.102`
- **Result:** 0% packet loss, confirming the target is "Up."



Step 2: Service Enumeration (Nmap)

I used **Nmap** to identify open ports and services on the target machine.

- **Command:** `nmap -sV 192.168.56.102`
- **Findings:** Identified open ports 21 (FTP), 22 (SSH), and 80 (HTTP).



---

4. Cryptographic Proof (The "Integrity" Pillar)

To demonstrate the **Integrity** of the technical documentation, a SHA-256 hash was generated for the project's cheat sheet. This ensures that the file has not been altered during the submission process.

- **Command:** `sha256sum Linux-Cheat-Sheet.md`

---

5. Conclusion

The environment is fully operational and secured within a private subnet. All baseline requirements for Task 1, including lab setup, networking basics, and tool familiarization, have been met. The environment is now ready for **Task 2: Reconnaissance and Scanning.**